Area of Parallelograms

Name _____ **Date** _____ Calculator allowed Version 1

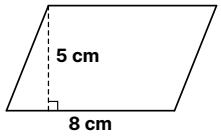
KEY WORDS base · perpendicular height · slant side · rhombus · diagonal · expression

A Base and perpendicular height ●○○ — ANSWER ON THE SHEET

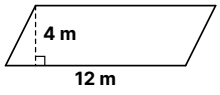
Remember: Area = base × perpendicular height. The height is the distance between the two parallel sides, at right angles to them.

1. Find the area of each parallelogram.

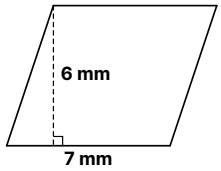
a)



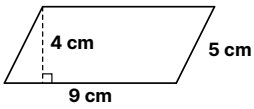
b)



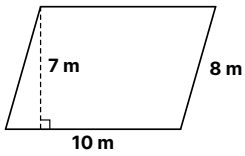
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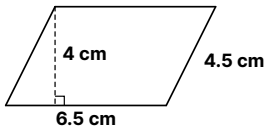
d)



e)



f)



B Finding a missing length ●●○ — WORK IN YOUR BOOK

Remember: Put what you know into Area = base × height and solve.

2. Find the missing length.

a) A parallelogram has an area of 48 cm² and a base of 8 cm. Find its perpendicular height.

b) A parallelogram has an area of 63 m² and a perpendicular height of 7 m. Find its base.

c) A parallelogram has an area of 90 cm² and a base of 12 cm. Find its perpendicular height.

d) A parallelogram has an area of 76 mm² and a perpendicular height of 8 mm. Find its base.

e) A parallelogram has an area of 42 m² and a base of 6 m. Find its perpendicular height.

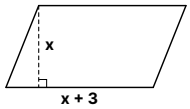
f) A parallelogram has an area of 117 cm² and a perpendicular height of 9 cm. Find its base.

C Lengths with algebra ●●● — WORK IN YOUR BOOK

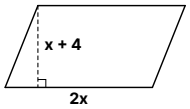
Remember: Write base × height with the expressions, then expand. To find x from a given area, write the equation and solve it.

3. Write an expression for the area of each parallelogram. Expand and simplify it.

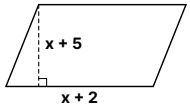
a)



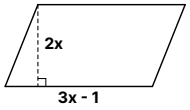
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c)



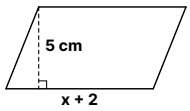
d)



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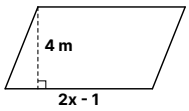
4. The area of each parallelogram is given. Write an equation and solve it to find x.

a)



The area is 40 cm². Find x.

b)



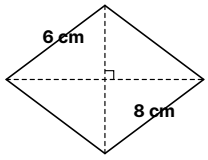
The area is 36 m². Find x.

D Rhombuses and kites ●●● WORK IN YOUR BOOK

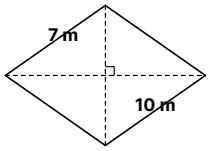
Remember: A rhombus or a kite can be cut by its diagonals into triangles. Area = $\frac{1}{2}$ × one diagonal × the other diagonal.

5. Find the area of each shape from its diagonals.

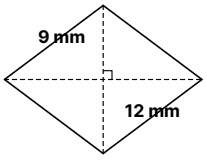
a)



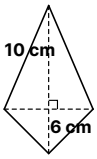
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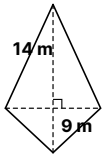
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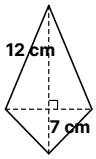
d)



e)



f)



E Problems ●●● WORK IN YOUR BOOK

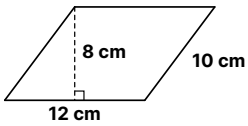
6. A parallelogram-shaped tile has a base of 30 cm and a height of 20 cm. Find the area of 240 tiles, in square metres.

7. A paddock is a parallelogram with a base of 85 m and a perpendicular height of 60 m. Fertiliser is spread at 1,250 kg per hectare (1 hectare = 10 000 m²). How much fertiliser is needed?

8. A logo is a parallelogram with a base of 6 cm and a height of 4 cm. It is enlarged by a scale factor of 2.5. Find the area of the enlarged logo.

★ Extension FOR EARLY FINISHERS

9. A parallelogram has a base of 12 cm and slant sides of 10 cm. Its perpendicular height on the base is 8 cm.



a) Find its area.

b) The area is also (slant side) × (the perpendicular distance between the slant sides). Use this to find that distance.

c) Explain why a parallelogram with base 12 cm and slant side 10 cm cannot have an area of more than 120 cm².

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Area of Parallelograms

Name _____ **Date** _____ Calculator allowed **Version 2**

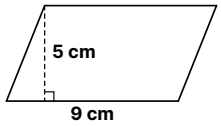
KEY WORDS base · perpendicular height · slant side · rhombus · diagonal · expression

A Base and perpendicular height ●○○ — ANSWER ON THE SHEET

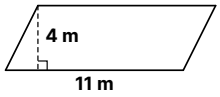
Remember: Area = base × perpendicular height. The height is the distance between the two parallel sides, at right angles to them.

1. Find the area of each parallelogram.

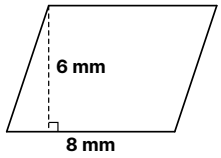
a)



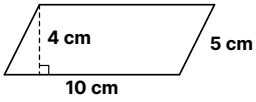
b)



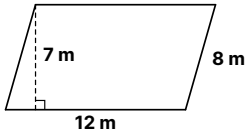
c)



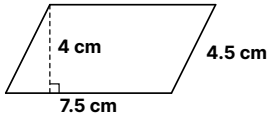
d)



e)



f)



B Finding a missing length ●●○ — WORK IN YOUR BOOK

Remember: Put what you know into Area = base × height and solve.

2. Find the missing length.

a) A parallelogram has an area of 54 cm² and a base of 9 cm. Find its perpendicular height.

b) A parallelogram has an area of 56 m² and a perpendicular height of 7 m. Find its base.

c) A parallelogram has an area of 84 cm² and a base of 12 cm. Find its perpendicular height.

d) A parallelogram has an area of 72 mm² and a perpendicular height of 8 mm. Find its base.

e) A parallelogram has an area of 45 m² and a base of 5 m. Find its perpendicular height.

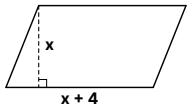
f) A parallelogram has an area of 126 cm² and a perpendicular height of 9 cm. Find its base.

C Lengths with algebra ●●● — WORK IN YOUR BOOK

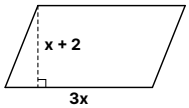
Remember: Write base × height with the expressions, then expand. To find x from a given area, write the equation and solve it.

3. Write an expression for the area of each parallelogram. Expand and simplify it.

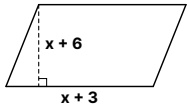
a)



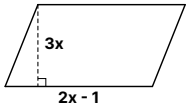
b)



c)



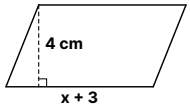
d)



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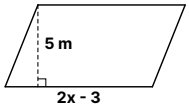
4. The area of each parallelogram is given. Write an equation and solve it to find x.

a)



The area is 40 cm². Find x.

b)



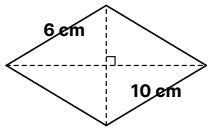
The area is 45 m². Find x.

D Rhombuses and kites ●●● WORK IN YOUR BOOK

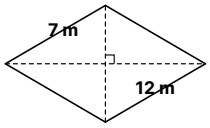
Remember: A rhombus or a kite can be cut by its diagonals into triangles. Area = $\frac{1}{2}$ × one diagonal × the other diagonal.

5. Find the area of each shape from its diagonals.

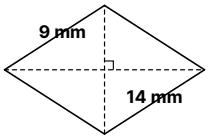
a)



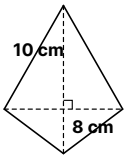
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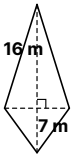
c)



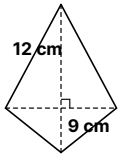
d)



e)



f)



E Problems ●●● WORK IN YOUR BOOK

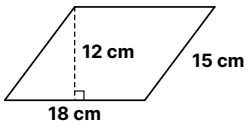
6. A parallelogram-shaped tile has a base of 25 cm and a height of 20 cm. Find the area of 300 tiles, in square metres.

7. A paddock is a parallelogram with a base of 90 m and a perpendicular height of 50 m. Fertiliser is spread at 1,200 kg per hectare (1 hectare = 10 000 m²). How much fertiliser is needed?

8. A logo is a parallelogram with a base of 8 cm and a height of 5 cm. It is enlarged by a scale factor of 3. Find the area of the enlarged logo.

Extension FOR EARLY FINISHERS

9. A parallelogram has a base of 18 cm and slant sides of 15 cm. Its perpendicular height on the base is 12 cm.



a) Find its area.

b) The area is also (slant side) × (the perpendicular distance between the slant sides). Use this to find that distance.

c) Explain why a parallelogram with base 18 cm and slant side 15 cm cannot have an area of more than 270 cm².

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